1. İki portlu bir devre elemanı için S-parametre matrisi $S_{11}=0.25$, $S_{12}=0$, $S_{21}=1.2$ ve $S_{22}=0.5$ olarak verilmektedir. Bu devre elemanının girişine 50Ω empedanslı 1mW güç sağlayan bir kaynak bağlanmakta, çıkışına ise 50Ω'luk bir yük bağlanmaktadır.



   a) Bu iki portlu devre elemanı resiprok mudur?
   b) Devre elemanının geri dönüş kaybını ve kazancını hesaplayınız.
   c) Girişten geri yansıyan gücü hesaplayınız.
   d) Yüke aktarılan gücü hesaplayınız.

$$S = \begin{pmatrix} 0,25 & 0 \\ 1,2 & 0,5 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix}$$

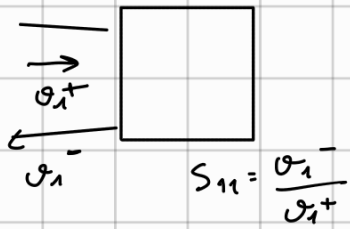
a) Resiprok olması için $S_{12} = S_{21}$ olmalıdır. Bu yüzden bu devre elemanı resiprok değildir. Demekki bu aktif eleman

b) $S_{11} \rightarrow$ Return loss

$S_{22} \rightarrow$ Output Return loss

$S_{21} \rightarrow$ gain

$S_{12} \rightarrow$ Isolation



$$RL = 20 \log (S_{11}^{-1})$$

(4,97) kitap

$$RL = -20 \log |S_{11}| \, dB = -20 \log (0,25) = -20(-0,602) = 12,04 \, dB$$

$$G = 10 \log_{10} |S_{21}|^2 = 20 \log |S_{21}| = 20 \log (1,2) = 20 \times 0,079 = 1,58 \, dB$$

c) $P_{in} = 1 mW$ $\quad P_{in_{dB}} = 10 \log \left(\frac{1mW}{10^{-3}} \right) = 0 dBm$

$$P_{ref_{dB}} = P_{in_{dB}} - RL_{dB} = 0 dBm - 12 dB = -12 dB$$



$$\frac{P_{in}}{R_L} = P_{ref}$$

d) $P_{out_{dB}} = P_{in_{dB}} + G_{dB} = 0 + 1,58 \, dB = 1,58 dBm$

12. The front end of a receiver for a cellular phone has a bandpass filter with a 25 MHz passband and a loss of 2 dB and is followed by two amplifier stages. The first stage has a gain of 20 dB and a noise figure of 0.5 dB and the second stage has a gain of 60 dB and a noise figure of 2 dB.

(a) Sketch the block diagram of the system as described.

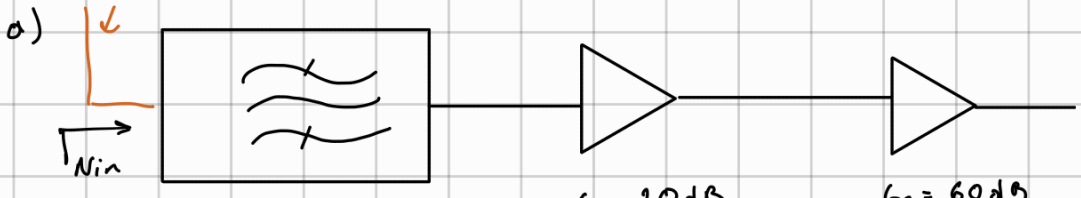
(b) What is the gain of the system?

(c) What is the noise figure of the bandpass filter?

(d) What is the noise figure of the system?

(e) The system is now connected to an antenna with an effective noise temperature of 30 K and which delivers a signal of 10 pW to the bandpass filter. Determine the noise temperature at the output of the system and hence the noise power in the 25 MHz bandwidth. Determine the signal-to-noise ratio at the output of the front-end system.

a)



b) The total gain of the system

$$G_{total\,(dB)} = G_{filter} + G_1 + G_2 = 78\ dB$$

c) $NF_{BPF} = L_{dB} = -G_{dB} = 2\ dB$ because $F_{BPF} = L = \frac{1}{G}$

d) $$F_{total} = F_{filter} + \frac{F_1 - 1}{G_{filter}} + \frac{F_2 - 1}{(G_{filter})\,G_1}$$

$F_{filter} = 10^{2/10} = 1,58$

$F_1 = 10^{0,5/10} = 1,122$, $F_2 = 10^{2/10} = 1,58$

$$\Rightarrow F_{total} = 1,58 + \frac{1,122-1}{0,631} + \frac{1,58-1}{0,63 \cdot 100} = 1,783$$

$$NF_{total} = 10\log(F_{total}) = 2,51\ dB$$

e) Noise temperature at the output

Tout

$SNR_{out} = \frac{P_{s,out}}{P_{n,out}}$

$N_{in} = kTB = (1{,}38 \cdot 10^{-23})(30)(25 \cdot 10^{6})$

($1{,}38 \cdot 10^{-23}$ ← $30k$)

$= 10{,}35 \text{ fW}$

$N_{out} = N_{in} \cdot F_{total} \cdot G_{total}$

$N_{out} = 1{,}19 \text{ mW}$

$T_{out} = \frac{N_{out}}{k \cdot B} =$

$S_{in} = 10 \text{ pW}$

$S_{out} = 10 \text{pW} \cdot G_{total} = 6{,}31 \cdot 10^{-4} \text{ W}$

$SNR_{out} = \frac{P_{s,out}}{P_{n,out}} = 595{,}5$

$SNR_{out} = 27{,}2 \text{ dB}$

$SNR_{in} = \frac{S_{in}}{N_{in}} = 966{,}2 = 29{,}85 \text{ dB}$

$SNR_{out(dB)} = SNR_{in(dB)} - NF_{total}$

$= 27{,}2 \text{ dB}$

13. A 75 Ω attenuator has a loss of 16 dB and is between a source with a Thevenin impedance of 75 Ω and a load of 75 Ω.

(a) What is the noise power, N_i, available from the 75 Ω source resistor at standard temperature (270 K) in a 1 MHz bandwidth?

(b) Now consider that the source is connected to the attenuator which is also connected to the load. If the source generates a modulated signal that is 1 MHz wide and has an available power, S_i, of 1 fW, what is SNR_i at the input to the attenuator at standard temperature?

(c) With the attenuator connected to the source, what is the Thevenin equivalent impedance looking into the output of the attenuator?

(d) Calculate the noise power, N_o, available from the attenuator with the source attached at standard temperature (270 K) in a 1 MHz bandwidth?

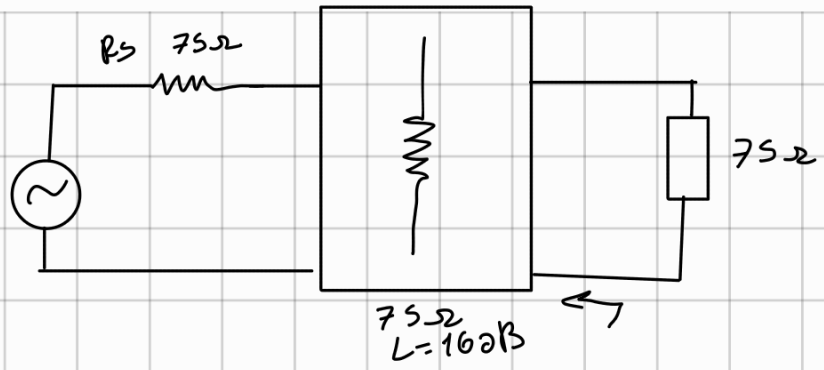
(e) What is the signal power, S_o, delivered to the load?

(f) What is the SNR at the load, SNR_o?

(g) What is the noise factor, F, of the attenuator?

(h) What is the noise figure, NF, of the attenuator?

a)



a şıkkında bağlamadan soruyor

$T = 270°K \quad R_s = 75\Omega$

$B = 1MHz$



$N_i = 4\,kTBR_s = 1,12\ pW$

b)

Input noise power $\frac{\overline{v^2_{in,n}}}{R_{in}} = \frac{kTR_s}{R_{in}} = kT$ in 1Hz BW

$= kTB$

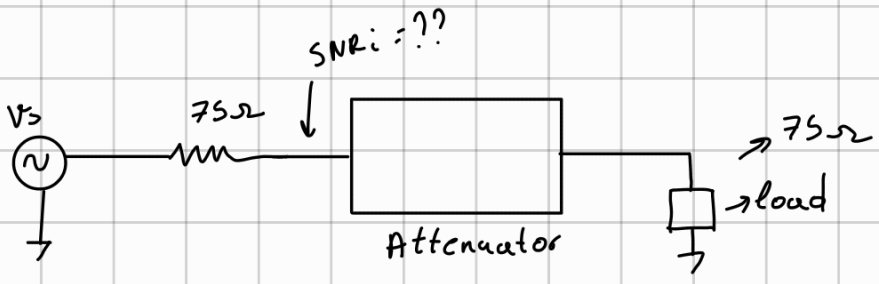
$k = 1,38 \cdot 10^{-23}\ J/K$

$T = 270K$

$B = 10^6\ Hz$

$N_i = kTB = 1,38 \cdot 10^{-23} \cdot 270 \cdot 10^6 = 3,726 \cdot 10^{-15} W$

$N_i = 3,726\ fW$



$SNR_i = \frac{S_i}{N_i} = \frac{1\ fW}{3,726\ fW} = 0,268$

$SNR_{i_{dB}} = 10\log(0,268) = -5,72\ dB$

c) Bir zayıflatıcı simetrik bağlantılara sahip olduğunda, giriş ve çıkış empedansları, kaynağın empandası ile aynıdır. Yani, zayıflatıcı 75Ω empedansa sahip olduğundan çıkış empedansı da 75Ω olacaktır.

Zayıflatıcının çıkış direncini buluruz

d) $G_{(dB)} = -L_{(dB)} = -16 dB$

$N_o = N_{in} \cdot F \cdot G$

$NF = L \rightarrow F = 10^{L/10}$

$G_{dB} = \frac{1}{L_{dB}} \rightarrow G = 10^{G_{dB}/10}$

e) $S_o = S_{in} \cdot G = 2,5,1 \cdot 10^{-18} W$

f) $SNR_o = \frac{S_o}{N_{out}} = 6,74 \cdot 10^{-3} \rightarrow SNR_{o(dB)} = -21,72$

g) $NF = L$

$F = 10^{L/10}$ ←

h) $NF = L$ ←